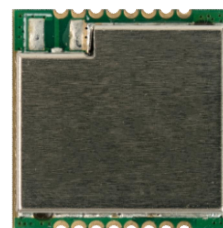
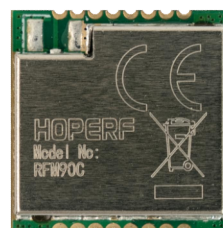
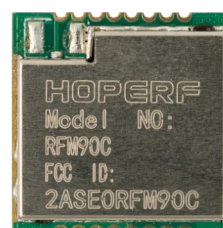


RFM90CW LoRa Transceiver Module

➤ General Description

RFM90CW sub-GHz radio transceivers are ideal for long range wireless applications. It is designed for long battery life with just 8mA of active receive current consumption. It can transmit up to +22dBm with highly efficient integrated power amplifiers. These devices support LoRa® modulation for LPWAN use cases and (G)FSK modulation for legacy use cases. The devices are highly configurable to meet different application requirements utilizing the global LoRaWAN™ standard or proprietary protocols. The devices are designed to comply with the physical layer requirements of the LoRaWAN™ specification released by the LoRa Alliance™. The radio is suitable for systems targeting compliance with radio regulations including but not limited to ETSI EN 300 220, FCC CFR 47 Part 15, China regulatory requirements and the Japanese ARIB T-108. Continuous frequency coverage from 150 MHz to 960 MHz allows the support of all major sub-GHz ISM bands around the world.

**RFM90CW-433****RFM90CW-868****RFM90CW-915**

➤ KEY PRODUCT FEATURES

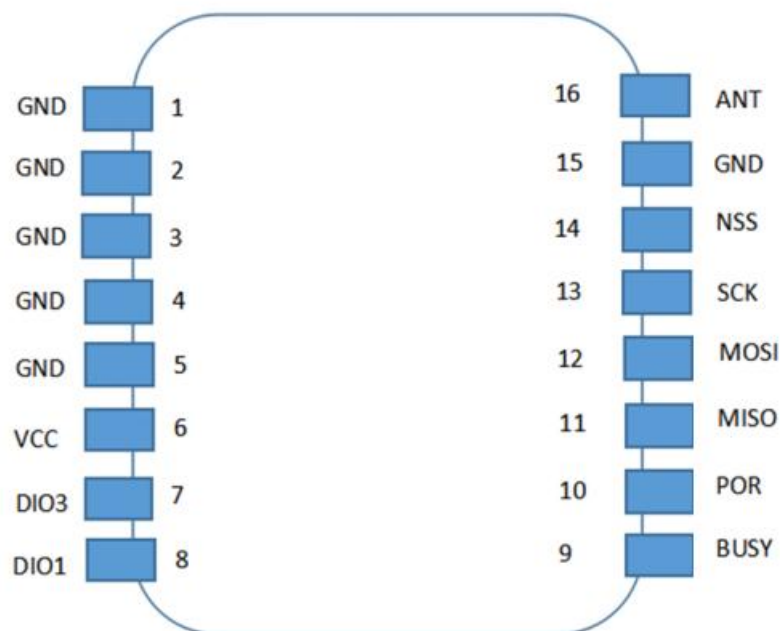
- ◆ (G)FSK, (G)MSK, LoRa™ Modem.
- ◆ +22dBm RF output .
- ◆ Programmable bit rate up to 300kbps(FSK)/62.5K(LORA).
- ◆ High sensitivity: down to -137dBm@LoRa BW 125KHz , SF12; -106dBm @FSK, 38.4kbps.
- ◆ Excellent blocking immunity.
- ◆ Low RX current of 8mA, 600 nA register retention.
- ◆ Fully integrated synthesizer with step 0.95 Hz.
- ◆ Built-in bit synchronizer for clock recovery.
- ◆ Preamble detection.
- ◆ 127dB Dynamic Range instantaneous/Package RSSI.
- ◆ Module Size: 16*16mm

➤ Applications

The level of integration and the low consumption within RFM90CW enable a new generation of Internet of Things applications.

- Smart meters
- Supply chain and logistics
- Building automation
- Agricultural sensors
- Smart cities
- Retail store sensors
- Asset tracking
- Street lights
- Environmental sensors
- Healthcare
- Safety and security sensors
- Remote control applications

➤ Pin Diagram



Picture 1: Pin Diagram (TOP View)

➤ Pin Description

NO.	Name	Description
1	ANT	RF signal output/input
2	GND	Ground
3	NSS	SPI slave Select
4	SCK	SPI clock
5	MOSI	SPI slave input
6	MISO	SPI slave output
7	POR	Reset
8	BUSY	Busy indicator
9	DIO1	Interrupt Signal output
10	DIO3	Interrupt Signal output/External XO power supply
11	VCC	Power supply
12	GND	Ground
13	GND	Ground
14	GND	Ground
15	GND	Ground
16	GND	Ground

➤ Electrical Characteristics

● Absolute Maximum Ratings

Symbol	Description	Min	Max	Unit
VDDmr	Supply Voltage	-0.5	3.9	V
Tmr	Temperature	-55	+125	°C

● Operating Range

Symbol	Descriptionnnn	Min	Max	Unit
VDD	Supply voltage	1.8	3.7	V
Temperature	Operational temperature range	-40	+85	°C
CL	Load capacitance on digital ports	-	20	pF

● Transmit Mode Specifications

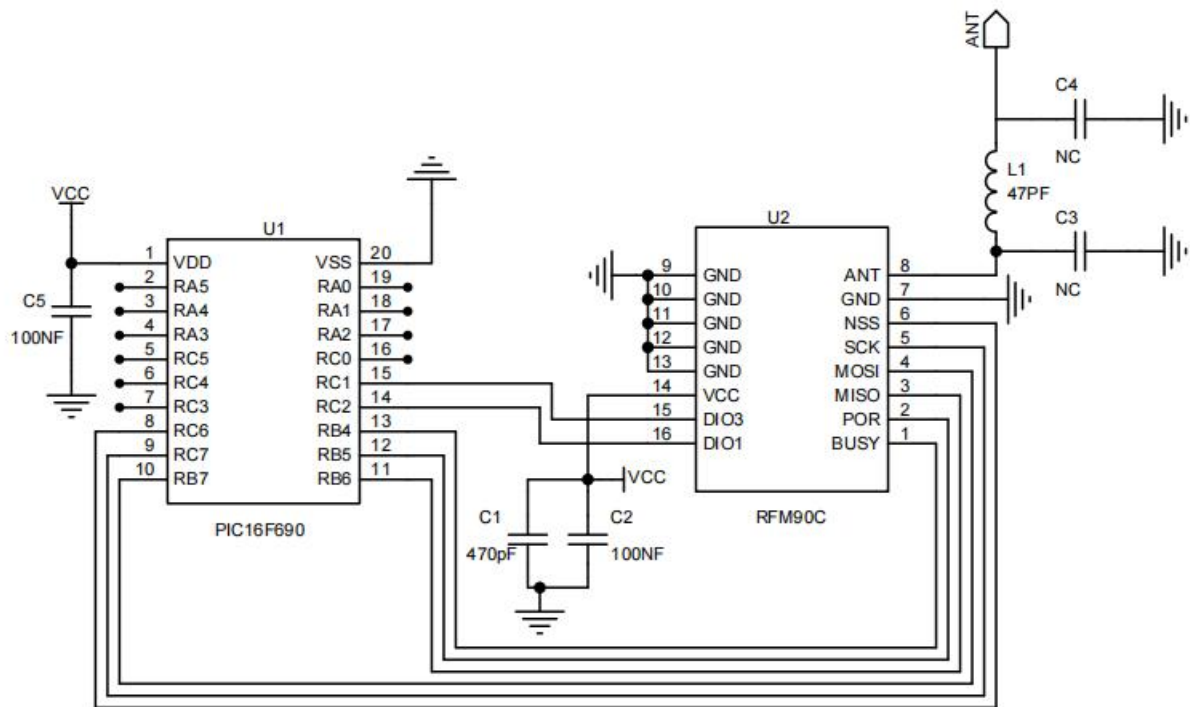
Specification	Condition	Min	Typical	Max	Unit
Frequency Range	433 MHz band	-	433.92	-	MHz
	868 MHz band	-	868	-	
	915 MHz band	-	915	-	

Tx Power	433 MHz band	-	22	-	dBm
	868 MHz band	-	22	-	
	915 MHz band	-	22	-	
Tx Drop	22dBm Vbat=2.7V	-	2	-	dB
	22dBm Vbat=2.4V	-	3	-	
	22dBm Vbat=1.8V	-	6	-	
IDDTX	433 MHz band	-	120	-	mA
	868 MHz band	-	145	-	
	915 MHz band	-	145	-	
Sleep Current	433 MHz band	-	0.6	-	uA
	868 MHz band	-	0.6	-	
	915 MHz band	-	0.6	-	

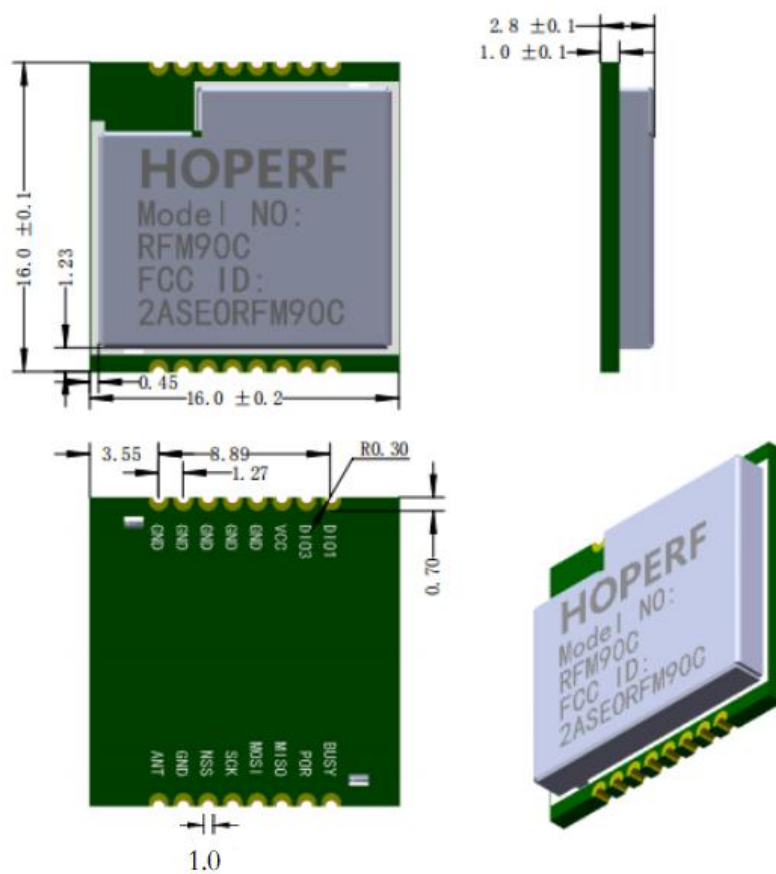
● **Receive Mode Specifications**

Specification	Condition	Min	Typical	Max	Unit
Sensitivity(FSK:) Rate=38.4kbps,FDA= 50KHz	433 MHz band	-	-108	-	dBm
	868 MHz band	-	-107	-	
	915 MHz band	-	-106	-	
Sensitivity(Lora) SF=12, BW=125KHz	433 MHz band	-	-137	-	dBm
	868 MHz band	-	-137	-	
	915 MHz band	-	-137	-	
IDDRX	FSK: Rate=38.4kbps	-	9.1	-	mA
	LoRa: SF=12, BW=125KHz	-	8.8	-	

➤ Reference schematic design



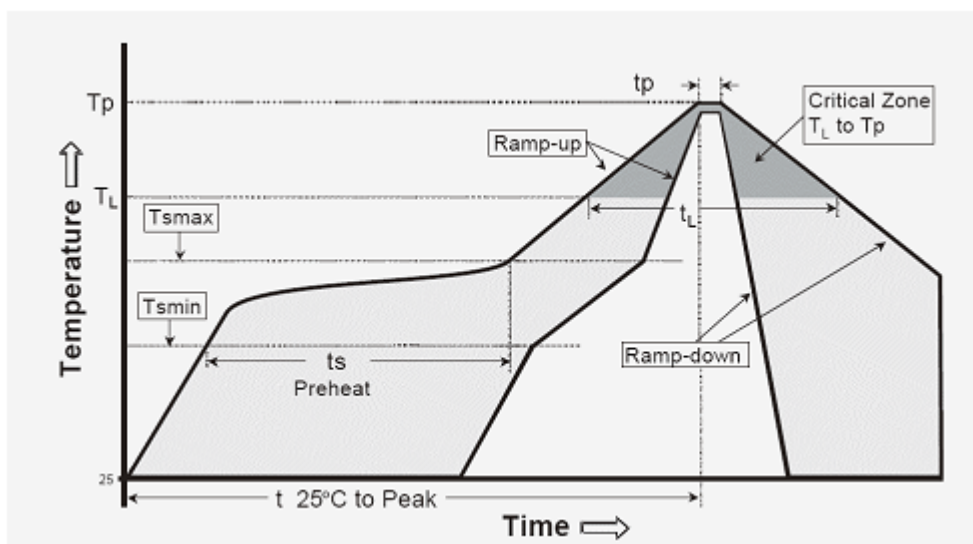
➤ Module Dimension



Picture 2: Module Dimension

➤ Reflow soldering temperature curve

It is recommended to comply with the related IPC standards and depend on the actual curve of the solder paste to setup the reflow soldering temperature curve.



Pre-heating temperature	
- Minimum temperature (Tsmmin)	150 °C
- Maximum temperature (Tsmmax)	200 °C
- Warm-up time (ts)	60 ~ 180 sec
Temperature ramping rate for preheating to the peak temperature (TL-TP)	3 °C/sec (maximum)
Soldering time	60 ~ 90 sec
- Soldering temperature	217 °C
- Time to reach above the melting point	60 ~ 150 sec
Peak temperature	240 +/- 5 °C
Soldering time for peak temperature +/- °C	10~30 sec
Cooling rate	6 °C/sec (maximum)
Full soldering time	8 min (maximum)

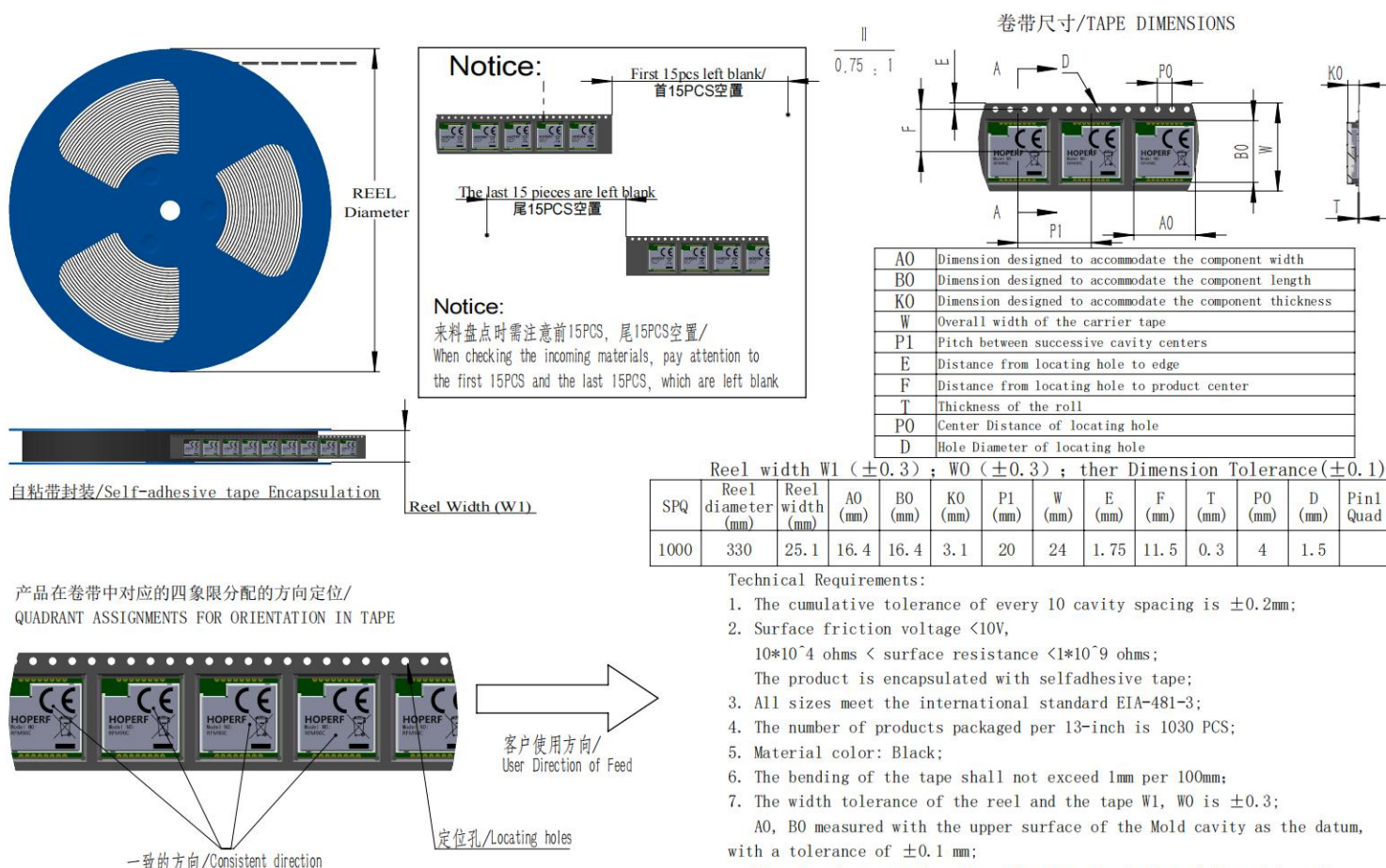
Notes

- The wettability of lead-free solder materials is worse than that of tin-lead eutectic materials, and the melting zone temperature is higher. To achieve better soldering results and protect the solder

joints from oxidation, use nitrogen protection during reflow soldering.

- For double-sided products, avoid reflow soldering with the module facing downward (that is, the module should be soldered during the second reflow soldering) to prevent the components on the module from falling off or shifting.
- The module needs to be fully tinned, and it is recommended to use a stepped steel mesh with a thickness of 0.15MM or above.
- When the module must be re-soldered, we recommend manual soldering to avoid possible damage to the module caused by multiple reflow soldering processes.
- Customers should develop welding processes that comply with the above tips based on actual production conditions.

➤ Packaging Information



➤ Ordering Information

Part No.	Working Frequency	Certifications
RFM90CW-433S2	433.92MHz	
RFM90CW-868S2	868MHz	CE(EU)
RFM90CW-915S2	915MHz	FCC(USA)

➤ Revision History

Version	Date	Modification
V1.0	2022.03.14	Initial version
V1.02	2025.02.20	Add the RFM90CW-433S2 and RFM90CW-868S2 information
V1.03	2026.03.17	Update reflow soldering temperature curve, reference designs, packaging information and certification information.

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